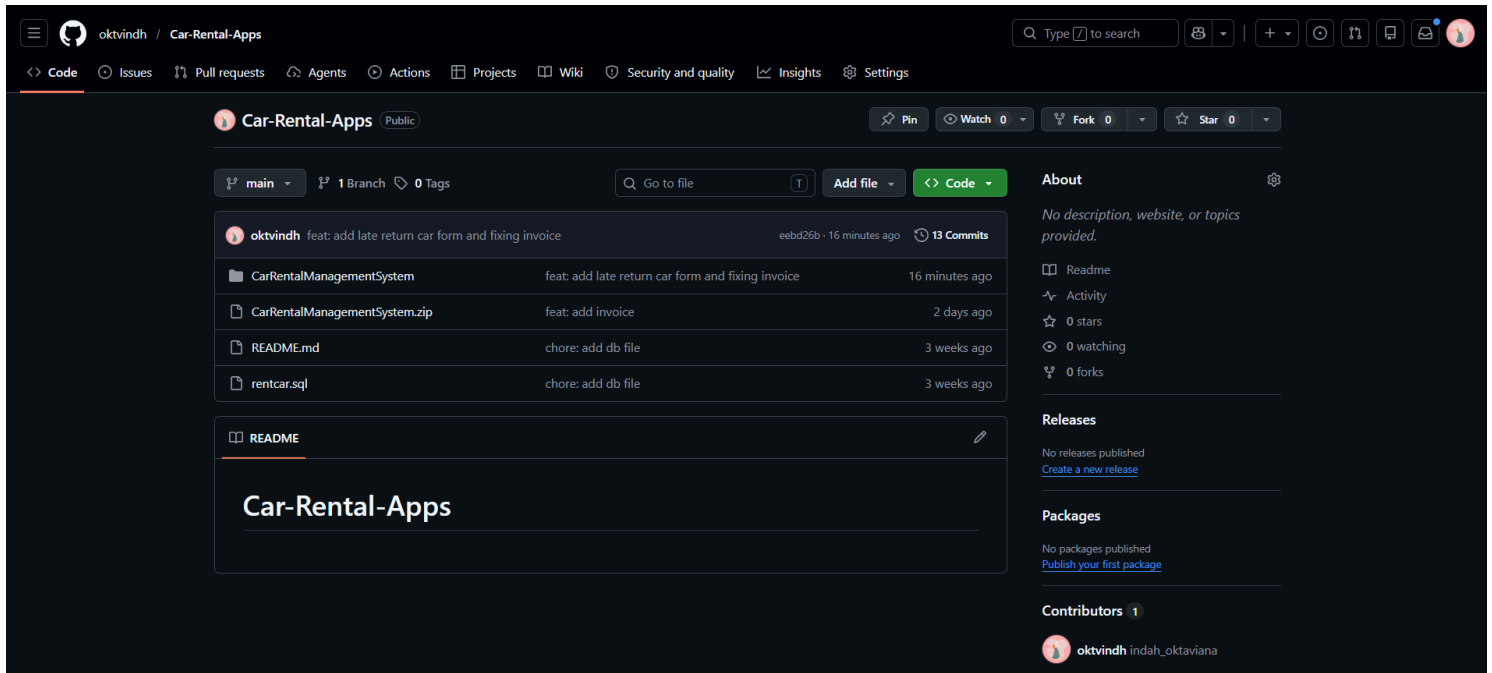


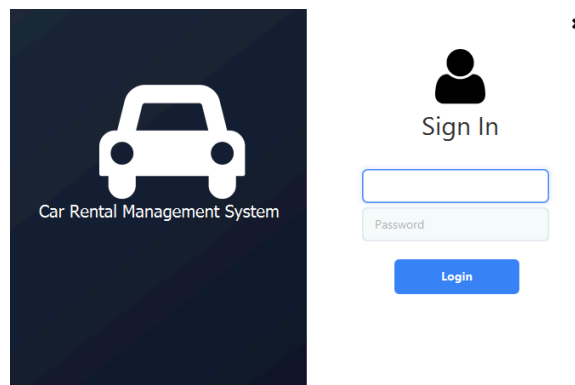
# LAPORAN APLIKASI CAR RENTAL MANAGEMENT SYSTEM

## Dokumentasi dan Timeline Pengerjaan

<https://github.com/oktvindh/Car-Rental-Apps>



## 1. Form Login



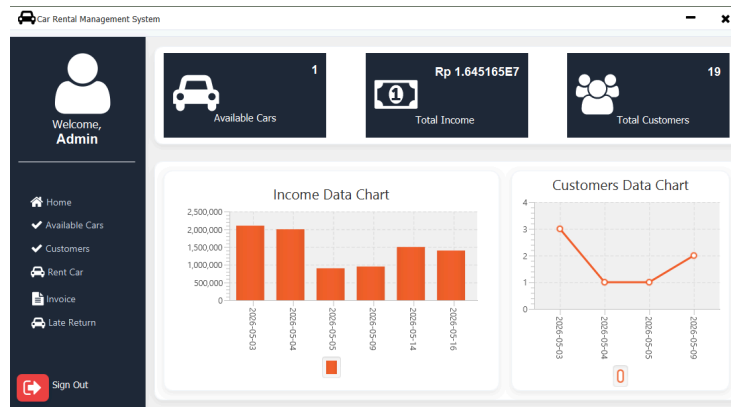
## Fungsi Form Login

- Memvalidasi akun pengguna.
- Membatasi akses aplikasi hanya untuk user yang memiliki akun.
- Menghubungkan aplikasi dengan database untuk pengecekan data login.

## Alur Login

1. User memasukkan username dan password.
2. Sistem memeriksa data pada database.
3. Jika data benar, user diarahkan ke Dashboard/Home.
4. Jika salah, sistem menampilkan pesan error.

## 2. Home



### Informasi yang Ditampilkan

- Total mobil tersedia.
- Total pendapatan rental.
- Total customer.
- Grafik pendapatan.
- Grafik data customer.

### Fungsi Home

- Menampilkan ringkasan data aplikasi.
- Mempermudah admin memonitor kondisi rental mobil.
- Menampilkan visualisasi data melalui chart.

## 3. Available Cars Form

The screenshot shows the Available Cars Form. It includes a sidebar with navigation links: Home, Available Cars, Customers, Rent Car, Invoice, Late Return, and Sign Out. The main area contains a form for adding or updating car information, with fields for Car ID, Brand, Model, Status, Price, and Plate Number. There are buttons for Update, Insert, Import, Clear, and Delete. Below the form is a table displaying the list of available cars.

| Car ID | Brand  | Model           | Plate Number |
|--------|--------|-----------------|--------------|
| 6      | Toyota | Avanza update   | F267YZ       |
| 7      | Honda  | Civic update    | B578XY       |
| 13     | Honda  | model 12 UPDATE | F123XY       |

### Fungsi Form

- Menambahkan data mobil.
- Mengubah data mobil.
- Menghapus data mobil.
- Menampilkan daftar mobil.
- Mengupload gambar mobil.
- Melakukan pencarian data mobil.

## Data Mobil yang Dikelola

- Car ID
- Brand
- Model
- Plate Number
- Price
- Status
- Image

## Fitur CRUD

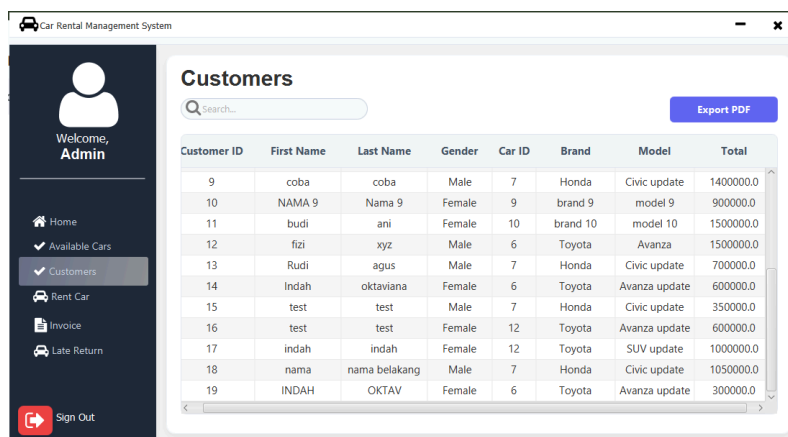
1. Insert  
Menambahkan mobil baru ke database.
2. Update  
Mengubah data mobil yang sudah ada.
3. Delete  
Menghapus data mobil dari database.
4. Clear  
Membersihkan input form.

## Status Mobil

- Available
- Not Available

**Tabel Mobil :** Menampilkan seluruh data mobil yang tersimpan pada database menggunakan TableView JavaFX.

## 4. Customers Form + Report



The screenshot shows a web application titled "Car Rental Management System". On the left is a dark sidebar with a user profile icon and the text "Welcome, Admin". Below this are menu items: "Home", "Available Cars", "Customers" (which is highlighted), "Rent Car", "Invoice", and "Late Return". At the bottom of the sidebar is a "Sign Out" button. The main content area is titled "Customers" and features a search bar and an "Export PDF" button. Below these is a table with the following data:

| Customer ID | First Name | Last Name     | Gender | Car ID | Brand    | Model         | Total     |
|-------------|------------|---------------|--------|--------|----------|---------------|-----------|
| 9           | coba       | coba          | Male   | 7      | Honda    | Civic update  | 1400000.0 |
| 10          | NAMA 9     | Nama 9        | Female | 9      | brand 9  | model 9       | 900000.0  |
| 11          | budi       | ani           | Female | 10     | brand 10 | model 10      | 1500000.0 |
| 12          | fizi       | xyz           | Male   | 6      | Toyota   | Avanza        | 1500000.0 |
| 13          | Rudi       | agus          | Male   | 7      | Honda    | Civic update  | 700000.0  |
| 14          | Indah      | oktaviana     | Female | 6      | Toyota   | Avanza update | 600000.0  |
| 15          | test       | test          | Male   | 7      | Honda    | Civic update  | 350000.0  |
| 16          | test       | test          | Female | 12     | Toyota   | Avanza update | 600000.0  |
| 17          | indah      | indah         | Female | 12     | Toyota   | SUV update    | 1000000.0 |
| 18          | nama       | nama belakang | Male   | 7      | Honda    | Civic update  | 1050000.0 |
| 19          | INDAH      | OKTAV         | Female | 6      | Toyota   | Avanza update | 300000.0  |

## Informasi Customer

- Customer ID
- First Name
- Last Name
- Gender

- Car ID
- Brand
- Model
- Total pembayaran

### Fungsi Form

- Menampilkan riwayat customer.
- Mencari data customer.
- Menampilkan transaksi customer.

**Customer Report** : menyediakan fitur export/print report customer menggunakan JasperReport.

### Isi Report Customer

- Data customer lengkap.
- Informasi mobil yang dirental.
- Total pembayaran customer.

### Fungsi Report

- Dokumentasi transaksi customer.
- Mempermudah admin melakukan pencetakan laporan.
- Membantu proses administrasi rental.

## 5. Rent Car Form

| Car ID | Brand  | Model           | Plate Number | Price (R) |
|--------|--------|-----------------|--------------|-----------|
| 6      | Toyota | Avanza update   | F267YZ       | 300000    |
| 7      | Honda  | Civic update    | B578XY       | 350000    |
| 13     | Honda  | model 12 UPDATE | F123XY       | 350000    |

### Alur Rental Mobil

1. Admin memilih mobil.
2. Admin mengisi data customer.
3. Admin menentukan tanggal rental.
4. Sistem menghitung total biaya rental.
5. Customer melakukan pembayaran.
6. Sistem menghitung balance/kembalian.
7. Data transaksi disimpan ke database.

### Data yang Digunakan

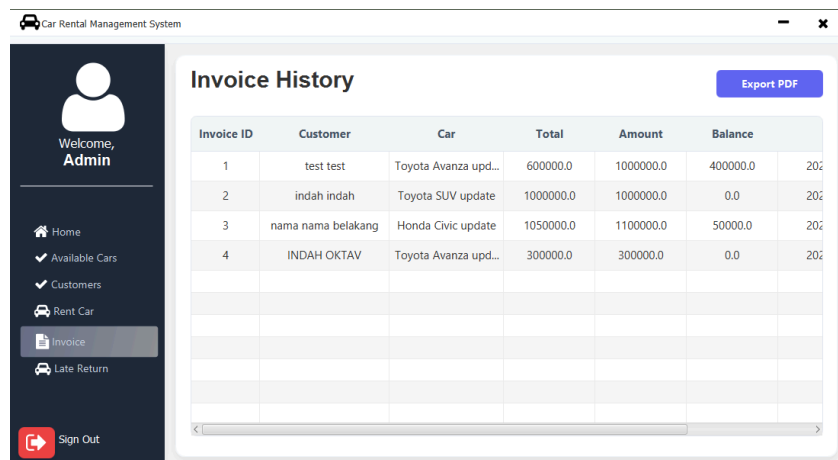
- Car ID
- Brand
- Model
- First Name
- Last Name
- Gender
- Date Rented
- Date Returned
- Total
- Amount
- Balance

### Fitur Perhitungan

- Perhitungan total harga berdasarkan lama rental.
- Perhitungan uang pembayaran.
- Perhitungan uang kembalian.

**Tabel Available Cars :** Pada bagian bawah form terdapat tabel mobil tersedia yang membantu admin melihat daftar mobil yang dapat dirental.

## 6. Invoice + Invoice History Report + Invoice per Customer



The screenshot shows a web application titled "Car Rental Management System". On the left is a dark sidebar with a user profile icon and the text "Welcome, Admin". Below this are navigation links: Home, Available Cars, Customers, Rent Car, Invoice (highlighted), and Late Return. At the bottom of the sidebar is a "Sign Out" button. The main content area is titled "Invoice History" and has an "Export PDF" button. It contains a table with the following data:

| Invoice ID | Customer           | Car                  | Total     | Amount    | Balance  |     |
|------------|--------------------|----------------------|-----------|-----------|----------|-----|
| 1          | test test          | Toyota Avanza upd... | 600000.0  | 1000000.0 | 400000.0 | 20% |
| 2          | indah indah        | Toyota SUV update    | 1000000.0 | 1000000.0 | 0.0      | 20% |
| 3          | nama nama belakang | Honda Civic update   | 1050000.0 | 1100000.0 | 50000.0  | 20% |
| 4          | INDAH OKTAV        | Toyota Avanza upd... | 300000.0  | 300000.0  | 0.0      | 20% |
|            |                    |                      |           |           |          |     |
|            |                    |                      |           |           |          |     |
|            |                    |                      |           |           |          |     |
|            |                    |                      |           |           |          |     |
|            |                    |                      |           |           |          |     |
|            |                    |                      |           |           |          |     |
|            |                    |                      |           |           |          |     |
|            |                    |                      |           |           |          |     |

### A. Invoice Form

Form Invoice digunakan untuk menampilkan histori transaksi rental yang telah dilakukan.

### Informasi Invoice

- Invoice ID
- Customer
- Car
- Total
- Amount
- Balance
- Date

### Fungsi Invoice

- Menyimpan histori transaksi rental.
- Menampilkan data pembayaran customer.
- Mempermudah pencarian transaksi.

## B. Invoice History Report

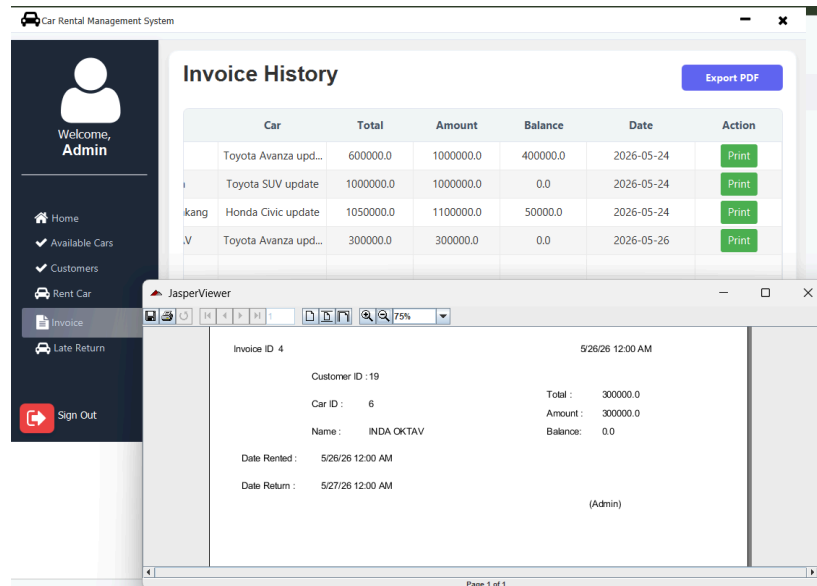
### Isi Report

- Seluruh data invoice rental.
- Informasi pembayaran customer.
- Tanggal transaksi.

### Fungsi Report

- Arsip transaksi rental.
- Bukti pembayaran.
- Dokumentasi administrasi.

## C. Invoice per Customer



Fitur Invoice per Customer digunakan untuk menampilkan invoice khusus berdasarkan customer tertentu.

### Fungsi

- Menampilkan detail transaksi individual customer.
- Digunakan sebagai bukti pembayaran customer.
- Mempermudah pencetakan invoice per transaksi.

### Informasi yang Ditampilkan

- Nama customer
- Mobil yang dirental
- Total pembayaran
- Amount
- Balance

- Tanggal rental

## 7. Late Return Form + Late Return Report

### A. Late Return Form

The screenshot shows the 'Car Rental Management System' interface. On the left is a dark sidebar with a user profile 'Welcome, Admin' and a menu including Home, Available Cars, Customers, Rent Car, Invoice, Late Return (highlighted), and Sign Out. The main content area is divided into two sections. The top section is the 'Late Return Form' with input fields for Customer ID, Brand, Model, Late Days, Fine (set to Rp 0), Amount, and Date Returned. It also shows a 'Balance' of Rp 0 and a 'Save' button. The bottom section is the 'Late Return History' table, which includes a 'Print Report' button and a table with 7 columns: Customer ID, Brand, Model, Late Days, Fine, Amount, and Balance. The table contains 4 rows of data.

| Customer ID | Brand  | Model        | Late Days | Fine     | Amount   | Balance |
|-------------|--------|--------------|-----------|----------|----------|---------|
| 2           | Toyota | Innova       | 3         | 0.0      | 0.0      | 0.0     |
| 18          | Honda  | Civic update | 3         | 0.0      | 0.0      | 0.0     |
| 5           | Toyota | Innova       | 3         | 150000.0 | 200000.0 | 50000.0 |
| 4           | Toyota | Innova New   | 3         | 150000.0 | 200000.0 | 50000.0 |

Form Late Return digunakan untuk mengelola data keterlambatan pengembalian mobil.

### Fungsi Form

- Menghitung denda keterlambatan.
- Menginput pembayaran denda.
- Menghitung balance/kembalian.
- Menyimpan data keterlambatan ke database.

### Data yang Digunakan

- Customer ID
- Brand
- Model
- Late Days
- Fine
- Amount
- Balance
- Date Returned

### Alur Perhitungan Denda

1. Admin memasukkan jumlah hari keterlambatan.
2. Sistem menghitung total denda otomatis.
3. Admin memasukkan nominal pembayaran.
4. Sistem menghitung balance/kembalian.
5. Data disimpan ke database.

### Rumus Denda

$$\text{Fine} = \text{Late Days} \times 50.000$$

## **Tabel Late Return**

Menampilkan seluruh histori keterlambatan customer menggunakan TableView.

## **B. Late Return Report**

Aplikasi menyediakan fitur report keterlambatan menggunakan JasperReport.

### **Isi Report**

- Customer ID
- Brand
- Model
- Late Days
- Fine
- Amount
- Balance
- Date Returned

### **Fungsi Report**

- Dokumentasi keterlambatan pengembalian mobil.
- Rekap data denda customer.
- Membantu administrasi dan monitoring keterlambatan.